

Linear Search

- sequential searching algorithm in C, that is used to find an element in list.
- compares each element of list with key till element is found or reach the end of list.

EX: Input: arr = {10, 50, 30, 70, 80, 60, 20, 90, 40},
key: 30

Output: key found at
index = 2

Explanation: Start from index 0, compare each element with key (30). When index 2 is reached, the element (30) matches the target value.

To search for given element using linear search, follow the approach:

1. from start of dataset
2. compare the current element with key
3. If element is equal to key, return index
4. else, increment the index & repeat the step 2 & 3.
5. If it reach end of list without finding the element equal to key, return some value to represent that element is not found.

implement linear search:

```
#include <stdio.h>
int linearsearch(int *arr,
                int n, int
                key)
```

```
{
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            return i;
        }
    }
}
```

```
3
3
else
    return -1;
}
```

```
int main() {
    int arr[] = {10, 50, 30, 70, 80,
                60, 20, 90, 40};
    int n = sizeof(arr) / sizeof
              (arr[0]);
```

```
int key = 30;
```

```
int i = linearsearch(arr, n, key);
```

```
if (i == -1)
    printf("Key not found");
```

```
else
    printf("Key found at
           index: %d", i);
```

```
return 0;
```

```
}
```

output:

key found at index: 2

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